Autonomous Next-Generation Aerial Refueling System (ANGARS)

Requirements Analysis & CONOPS Report

Student Name: Joe Cohen

Mentor Name: Professor Shannon Dubicki

Instructor: C.J. Utara

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1 Report Introduction

1.1 Overview

This report intends to define the baseline requirements and CONOPS for the Autonomous Next-Generation Aerial Refueling System (ANGARS), a system solution designed to address critical capability gaps in todays worlds aerial refueling operations. Traditional refueling methods rely primarily on human operators, posing risks of human-induced errors such as fatigue, inefficiencies in contested environments, and incompatibilities with next-gen unmanned systems. ANGARS aims to revolutionize this medium through anonymously controlled, AI-driven systems with multiple sensors, tight security, and a scalable architecture. Through this report, we will analyze and decompose stakeholder needs into system requirements and behaviors that will be used to inform and constrain our design decisions.

1.2 Report Scope

This report begins with a needs analysis informed by stakeholder engagement and then goes into a structured breakdown of requirements to define what the system must do. Requirements are organized into operational, interface, functional, performance, and constraint categories to allow for the capture of the full scope of the system's contextual domain definition. These requirements address challenges posed by stakeholders, and the system needs to satisfy the needs of the proposals gap analysis and needs to be identified by stakeholders.

To wrap up the requirement analysis portion, we will discuss Key Performance Parameters (KPPs), which are mission-critical requirements. They directly relate to stakeholder priorities and key gaps the ANGARS system will address.

Moving into the CONOPS section, this report covers, at a high level, how the system works and how it is to be interacted with, along with an identification and description of the users/actors who will be directly using/operating the system. The report then breaks down three operational scenarios to provide context into critical mission functions, the environment in which the operation is occurring, and the specific actions the system takes to accomplish the mission scenario.

2 Requirements Analysis

2.1 Research Conducted

Northrop Grumman Aircraft Field/I&T Engineer

Research was conducted to understand how fixed-wing aircraft engineers who have designed, built, and seen requirements defined by similar customer domains would see capability needs from this system. Discussions with the NG I&T engineer emphasized the importance of redundant and fail-safe systems (N6) and facilitating a design with early consideration of testability (N11). These details were emphasized to address historical challenges in transitioning autonomous designs to flight fielded systems. Additionally, their experience with legacy and next-gen platforms informed the integration with tanker fleets (N14) need, stressing the need for backwards compatibilities to minimize the amount of cost and schedule required to implement newer systems, citing this as an important aspect for the adoption of any system. The engineer also advocated for real-time health monitoring (N8) with preemptive subsystem failure analysis, citing lessons learned from prior aircraft programs where signs of subsystems were apparent but no utilized in practice.

Navy Pilot/Engineer

Research was conducted to identify and understand actual users and how they might interact with the system and how this system would be helpful for them. Feedback from the Navy Pilot/Engineer emphasized challenges encountered during typical refueling operations, such as managing communications between them and ground centers. This information informed the mission coordination (N16) need to streamline schedule and status updates. They also went into detail about the importance of procedural checklists and standardized protocols utilized during training and real life flight scenarios, highlighting the importance of an intuitive Human Machine Interface (HMI) (N12) to reduce already complex operational complexity.

Independent Research

Independent research analyzed existing autonomous refueling prototypes, adversarial threat landscapes, and currently active government requests for this system. There is current demonstrated need from the industry. The Next-Generation Aerial Refueling System (NGAS/KC-Z) program explicitly priorities autonomous refueling for contested environments (N10) by 2030, with the Air Force identifying and defining it as a "critical deficiency" [GAO, 10]. There are a couple of companies that have begun to actively develop and design prototypes, the government and these companies have highlighted needs for autonomous refueling (N1) for unmanned and manned platforms, multi-platform compatibility (N2), operation in contested environments (N3), and highly secure data links (N4) – requirements mirrored in ANGARS' needs.

2.2 Stakeholder Needs

Stakeholders were polled to understand the current technology and capabilities in the field that ANGARS is attempting to impact. They were then polled to understand the needs in today's environments to begin a basis for desired capability. From there, we can take these pieces of information and start to form a concrete definition of the capability gap that the ANGARS system must satisfy. The table below highlights the definition of stakeholder needs at the time of requirements analysis. Stakeholder needs were recorded in the Cameo as <<extendedRequirements>> to ensure that the classification of the element was sufficiently important to convey that these elements are the foundation of the system architecture. To further emphasize the importance, Needs have an applied <<Capability>> stereotype for easy reference to the capabilities that must be achieved through system design and definition to ensure stakeholder satisfaction.

Table 1: Stakeholder Needs

#	Id	Name	Text	Applied Stereotype
1	N1	N1 Autonomous Refueling	Enable refueling operations without human intervention to reduce error risks.	extendedRequirement [Class] Capability [Class]
2	N2	N2 Multi-Platform Compatibility	Unmanned and legacy manned aircraft refueling support	extendedRequirement [Class] Capability [Class]
3	N3	N3 Contested Environment Operation	Maintain functionality in GPS denied, electronic warfare contested or adverse weather.	extendedRequirement [Class] Capability [Class]
4	N4	N4 Secure Communication	Ensure reliable and tamper-proof communication between tanker and receiver.	extendedRequirement [Class] Capability [Class]
5	N5	N5 Collision Avoidance	Detect and avoid potential collisions to prevent mid air accidents.	extendedRequirement [Class] Capability [Class]
6	N6	N6 Redundant Fail-Safe Systems	Incorporate multilayered redundancy for critical subsystems.	extendedRequirement [Class] Capability [Class]
7	N7	N7 Mission Reliability	Maintain ≥95% mission success rate when operating in nominal conditions.	extendedRequirement [Class] Capability [Class]
8	N8	N8 Health Monitoring & Diagnostics	For critical subsystems, maintain real time monitoring and predictive fault isolation	extendedRequirement [Class] Capability [Class]
9	N9	N9 Regulatory Compliance	Meet airworthiness, cybersecurity and safety standards as per authority guidelines	extendedRequirement [Class] Capability [Class]
10	N10	N10 Threat Detection & Response	Real time identification and mitigation of electronic warfare, jamming or airborne threats	extendedRequirement [Class] Capability [Class]
11	N11	N11 Testability & Verification	Support standardized and validated test protocols for autonomous operations and sensor performance.	extendedRequirement [Class] Capability [Class]
12	N12	N12 Human-Machine Interface (HMI)	Intuitive operator interfaces with manual override capability.	extendedRequirement [Class] Capability [Class]
13	N13	N13 Lifecycle Support	Enable efficient repair, upgrades, and sustainment over a 10-year service life.	extendedRequirement [Class] Capability [Class]
14	N14	N14 Integration with Tanker Platforms	Must be compatible with KC-46 and future tanker fleets with minimum alterations	extendedRequirement [Class] Capability [Class]
15	N15	N15 Operational Efficiency	Key performance metrics (for example, fuel transfer rate, downtime) should be tracked and reported	extendedRequirement [Class] Capability [Class]
16	N16	N16 Mission Coordination & Scheduling	Coordinate refueling missions with ground control, prioritize requests, generate schedules, and communicate status updates to aircraft and operators	extendedRequirement [Class] Capability [Class]

2.3 Requirements Organizational Rational

Requirements were broken into five categories.

Functional – Describes Defines **what** the system must do to achieve capabilities.

Operational – Describes how the system will be **used** in its environment.

Performance – Describes **measurable criteria** the system must achieve capabilities.

External Interface – Describes how the system is **interacted with** by external actors/stakeholders.

Constraint – Describes **limitations** on system design.

Using this range of specification types serves as a backbone to the requirements structure, emphasizing different contexts of the system specification. In sections below, requirements are organized by these requirement types, to make it easier for reader of the specification to understand specific contexts of the system specification.

Requirements were also sorted by their associated need as this is likely going to be important when verifying that stakeholder needs are met with the scope of context of all requirements types.

2.4 ANGARS Requirements

The process of deriving requirements for the ANGARS system was not a one-time event but a systematic and iterative approach. This ensured the requirements were traceable, feasible, and tightly aligned with the system's mission needs.

Once system needs are understood and identified, requirements can be derived based on achieving that need. Requirements derivation started with the need and then cycled through the various requirements categories to ensure that all contexts of the system needs were addressed. This process continued iteratively until left with a composite list of requirements sufficiently capturing the mission and capability needs of the ANGARS system to fulfill stakeholder desires.

Finally, to ensure traceability, requirements were linked to needs to ensure that all contexts of a need were satisfied and that intents were transparent to the derived requirement. Requirements could then be examined in a purely contextual domain (e.g., operational, interface, etc.) for further decomposition/design definition down the road.

The table below summarizes the count of each requirement classification (quantitative, binary, qualitative) at the time of initial requirement analysis derivation. This table will be maintained throughout the course of the capstone to understand the development of the requirement specification.

Table 2: Requirements Tracker

	Total	Quantitative	%	Binary	Qualitative
Requirements Analysis Report	154	48	31.1%	63	43
Functional Analysis Report					
Conceptual Design Report					
Trade Study					
Test Plan					
System Specifications					

2.4.1 Functional Requirements

Table 3: ANGARS Functional Requirements

#	Name	Derived From	classification	Text	Verify Method	KPP
1	ANGARS-11 Wingspan Compatibility	N2 Multi-Platform Compatibility	Binary	The system shall support refueling for autonomous aircraft with wingspans of 5m-15m (T), 5m-18m (O).	Test	false
2	ANGARS-12 Fuel Flow Adaptation	N2 Multi-Platform Compatibility	Qualitative	The system shall adjust fuel flow rate based on receiving aircraft type and properties.	Demonstration	false
3	ANGARS-13 Multi-Fuel Support	N2 Multi-Platform Compatibility	Quantitative	The system shall support ≥ 3 (T), 4 (O) fuel types (eg. Jet A, Jet A-1, Jet B, etc.).	Inspection	false
4	ANGARS-14 Fuel Capacity Check	N2 Multi-Platform Compatibility	Binary	The system shall verify receiving aircraft fuel capacity before transfer.	Demonstration	false
5	ANGARS-15 Altitude Boom Adjustment	N2 Multi-Platform Compatibility	Binary	The system shall automatically adjust the boom length to compensate for altitude differences in receiving aircraft.	Demonstration	false
6	ANGARS-16 Fuel Port Adaptation	N2 Multi-Platform Compatibility	Qualitative	The system shall automatically calibrate and adapt to account for variations in fuel port configurations on receiving aircraft.	Analysis	false
7	ANGARS-17 Altitude by Aircraft Type	N2 Multi-Platform Compatibility	Qualitative	The system shall automatically adjust its altitude based on receiving aircraft type.	Analysis	false
8	ANGARS-18 High-Altitude Refueling	N2 Multi-Platform Compatibility	Quantitative	The system shall support refueling at altitudes up to 35,000 feet (T), 40,000 feet (O).	Test	false
9	ANGARS-71 Pump Fault Detection	N8 Health Monitoring & Diagnostics	Binary	The system shall detect faults in fuel pumps within 5 seconds (T), 2 seconds (O).	Test	false
10	ANGARS-72 Predictive Pump Analysis	N8 Health Monitoring & Diagnostics	Qualitative	The system shall implement algorithms to predict pump failures using vibrational analysis.	Analysis	false
11	ANGARS-73 Fuel Temperature Monitoring	N8 Health Monitoring & Diagnostics	Quantitative	The system shall monitor fuel temperature within 2C (T), 1C (O) accuracy.	Test	false
12	ANGARS-74 Hydraulic Fault Speed	N8 Health Monitoring & Diagnostics	Quantitative	The system shall identify and diagnose hydraulic system faults in ≤ 10 seconds (T), 5 seconds (O).	Analysis	false
13	ANGARS-75 Sensor Isolation	N8 Health Monitoring & Diagnostics	Binary	The system shall isolate faulty sensors during diagnostics.	Demonstration	false
14	ANGARS-76 Low Pressure Alert	N8 Health Monitoring & Diagnostics	Binary	The system shall alert operators of low fuel pressure.	Demonstration	false
15	ANGARS-77 Log Archive	N8 Health Monitoring & Diagnostics	Binary	The system shall archive diagnostics logs for ≥ 1 year (T), 2 years (O).	Demonstration	false
16	ANGARS-103 HMI Mission Display	N12 Human-Machine Interface (HMI)	Qualitative	The HMI shall display real-time mission schedules, queued aircraft status, and threat alerts for operators.	Demonstration	false
17	ANGARS-104 Manual Override Speed	N12 Human-Machine Interface (HMI)	Quantitative	The HMI shall allow manual override within 2.5 seconds (T), 2 seconds (O).	Demonstration	false
18	ANGARS-105 Emergency Shutdown	N12 Human-Machine Interface (HMI)	Binary	The HMI shall include emergency shutdown controls.	Test	false
19	ANGARS-106 Tactile Alerts	N12 Human-Machine Interface (HMI)	Binary	The HMI shall provide tactile feedback for critical alerts.	Test	false
20	ANGARS-107 Voice Command Support	N12 Human-Machine Interface (HMI)	Qualitative	The HMI shall support voice commands for operator actions.	Test	false
21	ANGARS-108 HMI Refresh Rate	N12 Human-Machine Interface (HMI)	Quantitative	The HMI shall display graphical data at 30 Hz (T), 60 Hz (O).	Test	false
22	ANGARS-109 Subsystem Health Display	N12 Human-Machine Interface (HMI)	Qualitative	The HMI shall display subsystem health using red (fault), yellow (degraded), and green (nominal) per MIL-STD-1472G.	Inspection	false
23	ANGARS-110 Multilingual HMI	N12 Human-Machine Interface (HMI)	Qualitative	The HMI shall include the option for multilingual support for operators.	Demonstration	false
24	ANGARS-117 Refueling Queue Reprioritization	N12 Human-Machine Interface (HMI)	Qualitative	The HMI shall allow operators to manually reprioritize refueling queues during emergencies.	Demonstration	false
25	ANGARS-125 KC-46 Integration	N14 Integration with Tanker Platforms	Binary	The system shall integrate with existing KC-46 tankers without structural modifications.	Inspection	true
26	ANGARS-126 Future Tanker Compatibility	N14 Integration with Tanker Platforms	Qualitative	The system shall support software updates to maintain compatibility with future tanker models.	Inspection	false
27	ANGARS-127 MIL-STD-1553 Compliance	N14 Integration with Tanker Platforms	Binary	The system shall comply with MIL-STD-1553 for interfacing with tanker avionics to ensure common interfacing.	Inspection	false
28	ANGARS-128 Tanker Integration Time	N14 Integration with Tanker Platforms	Quantitative	The system shall maintain integration times with new tanker models within 48 hours (T), 36 hours (O).	Inspection	false
29	ANGARS-129 Tanker Power Compatibility	N14 Integration with Tanker Platforms	Binary	The system shall interface with tanker power supply outputs of 28VDC nominal.	Inspection	false
30	ANGARS-130 Tanker Fuel Validation	N14 Integration with Tanker Platforms	Binary	The system shall validate and log tanker fuel quantities before mission start.	Test	false
31	ANGARS-131 Tanker Waypoint Sync	N14 Integration with Tanker Platforms	Binary	The system shall synchronize its navigation data with tanker waypoints.	Inspection	false

2.4.2 Performance Requirements

Table 4: ANGARS Performance Requirements

#	Name		Derived From	classification	Text	Verify Method	KPP
1	ANGARS-44	Obstacle Detection Range	N5 Collision Avoidance	Quantitative	The system shall detect obstacles within 500m (T), 750m (O) with 99% accuracy.	Test	true
2	ANGARS-45	Collision Rerouting	N5 Collision Avoidance	Qualitative	The system shall automatically reroute flight paths to avoid collisions.	Analysis	false
3	ANGARS-46	Aircraft Separation	N5 Collision Avoidance	Quantitative	The system shall maintain ≥ 100m (T), 150m (O) separation from non-receiving aircraft.	Test	false
4	ANGARS-47	Collision Update Frequency	N5 Collision Avoidance	Quantitative	The system shall update collision risk assessments every 0.5 seconds.	Test	false
5	ANGARS-48	Collision Avoidance Priority	N5 Collision Avoidance	Qualitative	The system shall prioritize collision avoidance over mission timelines.	Analysis	false
6	ANGARS-49	Foreign Object Detection	N5 Collision Avoidance	Quantitative	The system shall detect birds/foreign objects ≥ 10cm (T), 7cm (O) wide.	Test	false
7	ANGARS-50	False Alert Limit	N5 Collision Avoidance	Quantitative	The system shall limit false collision alerts to no more than two (T) one (O) per mission.	Analysis	false
8	ANGARS-51	Predictive Obstacles Algorithms	N5 Collision Avoidance	Qualitative	The system shall employ predictive algorithms for obstacle trajectories and avoidance.	Analysis	false
9	ANGARS-52	Terrain Avoidance	N5 Collision Avoidance	Binary	The system shall avoid terrain obstacles located below 500m (T), 750m (O) above ground level (AGL)	Test	false
10	ANGARS-53	Collision Expectancy	N5 Collision Avoidance	Quantitative	The system shall have on average, a collision expectancy of no more than once every 1,000,000 mission hours (T) 1,500,000 (O) mission hours.	Analysis	true
11	ANGARS-86	Jamming Detection	N10 Threat Detection & Response	Quantitative	The system shall detect jamming signals within 10 seconds (T), 8 seconds (O) and transmit alerts to ground control.	Demonstration	false
12	ANGARS-87	Threat Countermeasures	N10 Threat Detection & Response	Binary	The system shall employ countermeasures in the event of airborne threats.	Test	false
13	ANGARS-88	Breach Containment	N10 Threat Detection & Response	Binary	The system shall detect and contain cybersecurity breaches by automatically disconnecting impacted systems.	Demonstration	false
14	ANGARS-89	GPS Spoof Detection	N10 Threat Detection & Response	Binary	The system shall detect spoofed GPS signals within 6 seconds (T), 2 seconds (O).	Test	false
15	ANGARS-90	Cyberattack System Disabling	N10 Threat Detection & Response	Binary	The system shall disable non-critical systems during cyberattacks to minimize impact from potentially compromised systems.	Test	false
16	ANGARS-91	Jamming Mitigation	N10 Threat Detection & Response	Quantitative	The system shall mitigate jamming attempts within 30 seconds (T), 15 seconds (O).	Test	true
17	ANGARS-92	Threat Uptime	N10 Threat Detection & Response	Quantitative	The system shall have 99% (T) 99.9% (O) uptime during threat conditions.	Analysis	false
18	ANGARS-93	Threat Logging	N10 Threat Detection & Response	Binary	The system shall log all threat responses for post-mission review.	Inspection	false
19	ANGARS-94	Countermeasure Updates	N10 Threat Detection & Response	Quantitative	The system shall update countermeasure protocols via over-the-air (OTA) updates within 24 hours (T), 12 hours (O) of new threat identification	Analysis	false
20	ANGARS-95	Threat Alert Priority	N10 Threat Detection & Response	Qualitative	The system shall prioritize threat alerts over non-critical communications during contested operations.	Analysis	false
21	ANGARS-139	Fuel Transfer Rate	N15 Operational Efficiency	Quantitative	The system shall transfer fuel at 1,200 gallons/minute (T), 3000 gallons/minute(O).	Test	false
22	ANGARS-140	Downtime Reporting Accuracy	N15 Operational Efficiency	Quantitative	The system shall report downtime metrics within 2% (T), 1% (O) error.	Inspection	false
23	ANGARS-141	Fuel Schedule Optimization	N15 Operational Efficiency	Qualitative	The system shall employ an optimized fuel transfering schedule based on receiving aircraft priority.	Analysis	false
24	ANGARS-142	Fuel Waste Limit	N15 Operational Efficiency	Quantitative	The system shall limit fuel waste to no more than 1.0% (T), 0.5% (O) of the total fuel consumed in each mission	Analysis	false
25	ANGARS-143	Fuel Transfer Accuracy	N15 Operational Efficiency	Quantitative	The system shall ensure that actual fuel transfer volumes are within 5.0% (T), 3.0% (O) of the planned values.	Test	false
26	ANGARS-144	Turbulence Minimization	N15 Operational Efficiency	Qualitative	The system shall adjust fuel flow to minimize receiving aircraft turbulence.	Analysis	false
27	ANGARS-145	Fuel Quality Check	N15 Operational Efficiency	Binary	The system shall verify fuel quality before transfer.	Inspection	false
28	ANGARS-146	Transfer Completion Reporting	N15 Operational Efficiency	Quantitative	The system shall report fuel transfer completion within 5 seconds (T), 10 seconds (O).	Demonstration	false
29	ANGARS-147	Low-Fuel Priority	N15 Operational Efficiency	Qualitative	The system shall prioritize refueling for low-fuel receiving aircraft first.	Analysis	false
30	ANGARS-148	Predictive Maintenance	N15 Operational Efficiency	Binary	The system shall automatically update its maintenance schedule by analyzing performance metrics to determine if maintenance is likely required	Demonstration	false

2.4.3 Interface Requirements

Table 5: ANGARS Interface Requirements

#	Name		Derived From	classification	Text	Verify Method	KPP
1	ANGARS-30	Pre-Comm Authentication	N4 Secure Communication	Binary	The system shall authenticate all receiving aircraft before data exchange.	Test	false
2	ANGARS-31	Low-Latency Commands	N4 Secure Communication	Quantitative	The system shall maintain ≤ 15ms (T), 10ms (O) latency in command transmission.	Test	true
3	ANGARS-32	Anti-Jamming Hopping	N4 Secure Communication	Binary	The system shall employ frequency-hopping techniques to counter jamming attempts.	Test	false
4	ANGARS-33	Communication Uptime	N4 Secure Communication	Quantitative	The system shall sustain communication uptime of ≥ 99.8% (T), 99.9% (O) between receiving aircraft.	Analysis	false
5	ANGARS-34	Data Loss Limit	N4 Secure Communication	Quantitative	The system shall have data packet loss of ≤ 0.2% (T), 0.1% (O).	Test	false
6	ANGARS-35	Unauthorized Connection Termination	N4 Secure Communication	Binary	The system shall terminate unauthenticated connections.	Test	false
7	ANGARS-36	RF Masking	N4 Secure Communication	Qualitative	The system shall employ RF signature masking techniques during covert operations.	Analysis	false
8	ANGARS-37	Certificate Validation	N4 Secure Communication	Binary	The system shall validate digital certificates for all receiving aircrafts.	Test	false
9	ANGARS-38	Message Integrity During Jamming	N4 Secure Communication	Quantitative	The system shall achieve ≥ 98% (T), 99% (O) message integrity during jamming.	Test	false
10	ANGARS-39	Zero-Trust Network	N4 Secure Communication	Qualitative	The system shall implement a zero-trust network architecture, verifying all access requests regardless of origin.	Analysis	false
11	ANGARS-40	Encryption Recovery	N4 Secure Communication	Quantitative	The system shall re-establish encrypted communication within 5 seconds (T), 4 seconds (O) of signal loss.	Test	false
12	ANGARS-41	Tampered Data Erase	N4 Secure Communication	Binary	The system shall erase sensitive data and notify ground control if tampering is detected.	Demonstration	false
13	ANGARS-42	Quantum Encryption	N4 Secure Communication	Binary	The system shall use quantum-resistant encryption algorithms.	Inspection	false
14	ANGARS-43	AES-256 Encryption	N3 Contested Environment Operation N4 Secure Communication N10 Threat Detection & Response	Binary	The system shall encrypt all communications using AES-256 encryption protocol.	Inspection	true
15	ANGARS-111	Operator Action Logging	N12 Human-Machine Interface (HMI)	Binary	The HMI shall log operator actions and system context for audit purposes.	Inspection	false
16	ANGARS-112	Haptic Emergency Alerts	N12 Human-Machine Interface (HMI)	Binary	The HMI shall provide haptic feedback to operators for emergency alerts.	Demonstration	false
17	ANGARS-113	HMI Update Speed	N12 Human-Machine Interface (HMI)	Quantitative	The HMI shall update status displays every 0.5 seconds (T), 0.2 seconds (O).	Test	false
18	ANGARS-114	Ground Dashboard Integration	N12 Human-Machine Interface (HMI)	Qualitative	The HMI shall integrate with ground control station dashboards.	Demonstration	false
19	ANGARS-115	Critical Control Lockdown	N12 Human-Machine Interface (HMI)	Binary	The HMI shall disable non-essential controls during critical emergencies.	Test	false
20	ANGARS-116	Manual Override Instructions	N12 Human-Machine Interface (HMI)	Qualitative	The HMI shall provide step-by-step instructions for manual override.	Analysis	false
21	ANGARS-132	Tanker Navigation Interface	N14 Integration with Tanker Platforms	Qualitative	The system shall interface with tanker navigation systems.	Inspection	false
22	ANGARS-133	Tanker Health Reporting	N14 Integration with Tanker Platforms	Qualitative	The system shall send tanker health data to ground stations.	Analysis	false
23	ANGARS-134	Link 16 Compliance	N14 Integration with Tanker Platforms	Binary	The system shall comply with Link 16 messaging protocols.	Inspection	false
24	ANGARS-135	Tanker Data Bus Auto-Detection	N14 Integration with Tanker Platforms	Binary	The system shall automatically detect and configure tanker-specific data buses.	Inspection	false
25	ANGARS-136	Fuel Data Sync	N14 Integration with Tanker Platforms	Qualitative	The system shall synchronize fuel data with tanker fuel data logistical systems.	Inspection	false
26	ANGARS-137	Tanker Comm Latency	N14 Integration with Tanker Platforms	Quantitative	The system shall have a communications interface latency of 5ms (T), 3ms (O) for critical commands.	Test	false

2.4.4 Operational Requirements

Table 6: ANGARS Operational Requirements

#	Name	Derived From	classification	Text	Verify Method	KPP
1	ANGARS-1 Autonomous Refueling	N1 Autonomous Refueling	Binary	The system shall execute autonomous refueling via onboard sensors without human assistance.	Demonstration	false
2	ANGARS-2 Refueling Time	N1 Autonomous Refueling	Quantitative	The system shall complete refueling within 10 minutes (T), 8 minutes (O) of handshake (alignment) with receiving aircraft.	Test	true
3	ANGARS-3 Proximity Safety Abort	N1 Autonomous Refueling	Binary	The system shall abort refueling if the receiving aircraft proximity exits established safe bounds.	Test	false
4	ANGARS-4 Aircraft ID Verification	N1 Autonomous Refueling	Binary	The system shall verify and validate receiving aircraft identity before beginning fuel transfer.	Demonstration	false
5	ANGARS-5 Turbulence Stability	N1 Autonomous Refueling	Binary	The system shall maintain a stable connection with the receiving aircraft during turbulence.	Test	false
6	ANGARS-6 Probe Alignment Accuracy	N1 Autonomous Refueling	Quantitative	The system shall align the refueling probe within 0.5m (T), 0.25m (O) of the receiving aircrafts fueling port.	Test	false
7	ANGARS-7 Fuel Leak Detection	N1 Autonomous Refueling	Binary	The system shall detect fuel leaks and cease fuel transfers immediately.	Test	false
8	ANGARS-8 Adaptive Fuel Speed	N1 Autonomous Refueling	Qualitative	The system shall adjust refueling speed based on receiving aircraft stability.	Analysis	false
9	ANGARS-9 Probe Retraction	N1 Autonomous Refueling	Binary	The system shall retract the refueling probe when refueling has completed	Demonstration	false
10	ANGARS-10 Multi-Platform Refueling	N1 Autonomous Refueling N2 Multi-Platform Compatibility	Qualitative	The system shall employ refueling procedures for both manned and unmanned aircraft.	Test	false
11	ANGARS-19 GPS-Denied Navigation	N3 Contested Environment Operation	Binary	The system shall operate in GPS-denied environments by automatically switching to alternate navigation methods.	Test	false
12	ANGARS-20 Electronic Warfare Operation	N3 Contested Environment Operation	Qualitative	The system shall maintain functionality during electronic warfare conditions.	Analysis	false
13	ANGARS-21 Adverse Weather Operation	N3 Contested Environment Operation	Binary	The system shall operate without degredation in operational capability in adverse weather conditions (rain, fog, winds ≤25 knots).	Demonstration	false
14	ANGARS-22 Inertial Navigation	N3 Contested Environment Operation	Qualitative	The system shall utilize inertial navigation when GPS is not available (GPS denial, outages).	Demonstration	false
15	ANGARS-23 Low-Visibility Sensors	N3 Contested Environment Operation	Binary	The system shall switch to optical sensors in low-visibility conditions.	Test	false
16	ANGARS-24 Mission Task Prioritization	N3 Contested Environment Operation	Qualitative	The system shall automatically prioritize mission-critical tasks over non-critical tasks during contested operations.	Analysis	false
17	ANGARS-25 Terrain Navigation	N3 Contested Environment Operation	Qualitative	The system shall employ terrain-based navigation in GPS-denied environments.	Demonstration	false
18	ANGARS-26 Extreme Temperature Range	N3 Contested Environment Operation	Quantitative	The system shall operate in temperature ranges of -35C to +50C (T), -40C to +55C (O).	Demonstration	false
19	ANGARS-27 Contested Positioning Accuracy	N3 Contested Environment Operation	Quantitative	The system shall maintain positional accuracy of ≤ 5m (T), 4m (O) in contested environments.	Test	true
20	ANGARS-28 Threat Zone Avoidance	N3 Contested Environment Operation	Qualitative	The system shall automatically reroute missions to avoid known threat zones.	Analysis	false
21	ANGARS-29 Satellite Relay	N3 Contested Environment Operation	Binary	The system shall reroute mission data through relay satellites if direct ground links are jammed.	Test	false
22	ANGARS-62 Mission Success Rate	N7 Mission Reliability	Quantitative	The system shall achieve ≥95% mission success rate under nominal conditions.	Analysis	true
23	ANGARS-63 Failure Logging	N7 Mission Reliability	Qualitative	The system shall log mission failures for post-mission diagnostics.	Inspection	false
24	ANGARS-64 Fault Recovery Time	N7 Mission Reliability	Quantitative	The system shall recover from faults and resume operations within 2 minutes (T), 1 minute (O).	Test	false
25	ANGARS-65 Mission Abort Limit	N7 Mission Reliability	Quantitative	The system shall limit mission aborts caused by subsystem failures to ≤ 3% (T), 2% (O).	Analysis	false
26	ANGARS-66 Data Archival	N7 Mission Reliability	Binary	The system shall archive mission data for ≥ 30 days (T), 45 days post-operation.	Inspection	false
27	ANGARS-67 Post-Mission Reports	N7 Mission Reliability	Qualitative	The system shall generate post-mission performance reports.	Inspection	false
28	ANGARS-68 Status Notifications	N7 Mission Reliability	Binary	The system shall notify operators of mission success/failure within 1 minute (T), 30 seconds (O).	Demonstration	false
29	ANGARS-69 False Abort Limit	N7 Mission Reliability	Quantitative	The system shall limit false-positive mission aborts to no more than 2% (T), 1% (O).	Analysis	false
30	ANGARS-70 Fuel Compatibility Check	N7 Mission Reliability	Binary	The system shall validate fuel compatibility between receiving aircraft before transfer.	Test	false
31	ANGARS-149 Refueling Request Processing	N16 Mission Coordination & Scheduling	Quantitative	The system shall receive, authenticate, and process refueling requests from ground control via satellite within 10 seconds (T), 8 seconds (O).	Test	false
32	ANGARS-150 Refueling Priority Logic	N16 Mission Coordination & Scheduling	Qualitative	The system shall prioritize refueling requests based on aircraft fuel urgency, mission criticality, and operational efficiency algorithms.	Analysis	false
33	ANGARS-151 Schedule Generation Speed	N16 Mission Coordination & Scheduling	Quantitative	The system shall generate a refueling schedule and broadcast it to queued aircraft and the ground within 30 seconds (T), 20 seconds (O) of mission assignment.	Demonstration	false
34	ANGARS-152 Dynamic Schedule Updates	N16 Mission Coordination & Scheduling	Qualitative	The system shall update schedules dynamically in response to new requests or mission disruptions (e.g., threats, weather).	Test	false
35	ANGARS-153 Post-Mission Reporting	N16 Mission Coordination & Scheduling	Quantitative	The system shall generate and transmit post-mission reports (fuel transferred, anomalies, threats) to ground control within 5 minutes (T), 3 minutes (O) of mission completion.	Demonstration	false
36	ANGARS-154 Mission Status Updates	N16 Mission Coordination & Scheduling	Quantitative	The system shall transmit mission status updates to ground control every 30 seconds.	Demonstration	false

2.4.5 Constraint Requirements

Table 7: ANGARS Design Constraint Requirements

#	Name		Derived From	classification	Text	Verify Method	KPP
1	ANGARS-54	Redundant Pumps	N6 Redundant Fail-Safe Systems	Binary	The system shall have redundant fuel pumps.	Inspection	false
2	ANGARS-55	Power Loss Safe Mode	N6 Redundant Fail-Safe Systems	Binary	The system shall default to safe mode during power loss.	Test	false
3	ANGARS-56	Triple Redundancy	N6 Redundant Fail-Safe Systems	Binary	The system shall employ triple-redundant flight control systems.	Inspection	true
4	ANGARS-57	Backup Power	N6 Redundant Fail-Safe Systems	Binary	The system shall have backup power for critical subsystems.	Test	false
5	ANGARS-58	Hydraulic Isolation	N6 Redundant Fail-Safe Systems	Binary	The system shall automatically detect and isolate hydraulic failures to ensure faulting does not propegate throughout the rest of the system.	Demonstration	false
6	ANGARS-59	Dual Data Buses	N6 Redundant Fail-Safe Systems	Binary	The system shall employ dual-redundant data buses.	Inspection	false
7	ANGARS-60	Boom Fail-Safe Locks	N6 Redundant Fail-Safe Systems	Binary	The system shall include fail-safe mechanical locks for the refueling boom.	Inspection	false
8	ANGARS-61	Redundant IMUs	N6 Redundant Fail-Safe Systems	Binary	The system shall include redundant inertial measurement units (IMUs).	Test	false
9	ANGARS-78	FAA Compliance	N9 Regulatory Compliance	Qualitative	The system shall comply with FAA airworthiness standards.	Inspection	false
10	ANGARS-79	MIL-STD-882E Compliance	N9 Regulatory Compliance	Qualitative	The system shall meet or exceed MIL-STD-882E safety protocols.	Inspection	false
11	ANGARS-80	NATO Cybersecurity	N9 Regulatory Compliance	Qualitative	The system shall adhere to NATO cybersecurity standards.	Inspection	false
12	ANGARS-81	EMI Compliance	N9 Regulatory Compliance	Binary	The system shall pass electromagnetic interference (EMI) testing.	Inspection	false
13	ANGARS-82	STANAG 4406 Compliance	N9 Regulatory Compliance	Binary	The system shall comply with STANAG 4406 for secure messaging to all NATO platforms.	Test	false
14	ANGARS-83	Penetration Testing	N9 Regulatory Compliance	Binary	The system shall successfully pass all cybersecurity penetration tests with no exploitable vulnerabilities detected.	Test	false
15	ANGARS-84	ICAO Annex 6 Compliance	N9 Regulatory Compliance	Binary	The system shall meet or exceed ICAO Annex 6 safety requirements.	Test	false
16	ANGARS-85	DO-178C Compliance	N9 Regulatory Compliance	Binary	The system shall comply with DO-178C for software airworthiness.	Analysis	false
17	ANGARS-96	Annual Autonomy Testing	N11 Testability & Verification	Binary	The system shall meet all requirements of the autonomous mode test protocols each year.	Test	false
18	ANGARS-97	Sensor Accuracy Validation	N11 Testability & Verification	Quantitative	The system shall validate sensor accuracy during testing to ±1% (T), ±0.5% (O) during testing.	Test	false
19	ANGARS-98	Sandbox Software Testing	N11 Testability & Verification	Binary	The system shall validate all software updates in a sandbox environment before flight implementation.	Test	false
20	ANGARS-99	Algorithm Test Coverage	N11 Testability & Verification	Quantitative	The system shall ensure that all autonomous algorithms are thoroughly tested, achieving 98% (T), 100% (O) test coverage for autonomous algorithms against DO-178C standards and regulations.	Analysis	false
21	ANGARS-100	Sensor Calibration	N11 Testability & Verification	Quantitative	The system shall calibrate sensors to 0.5% (T), 0.25% (O) error margin semi-annually.	Demonstration	false
22	ANGARS-101	Software Checksum	N11 Testability & Verification	Binary	The system shall verify software integrity via checksum check pre-deployment.	Inspection	false
23	ANGARS-102	Hardware Durability Testing	N11 Testability & Verification	Quantitative	The system shall validate hardware durability over 8,000 (T) operational hours, 10,000 (O) operational hours.	Test	false
24	ANGARS-118	Component Replacement Time	N13 Lifecycle Support	Quantitative	The system shall support the replacement of critical components to restore operational capability within 4 hours (T), 2 hours (O).	Demonstration	false
25	ANGARS-119	Modular Subsystems	N13 Lifecycle Support	Qualitative	The system shall utilize modular subsystems to enable independent upgrades.	Inspection	false
26	ANGARS-120	Mean Repair Time	N13 Lifecycle Support	Quantitative	The system shall have a mean repair time (MTTR) of 2 hours (T), 1.5 hours (O).	Demonstration	false
27	ANGARS-121	Obsolescence Management	N13 Lifecycle Support	Qualitative	The system shall implement a 10-year obsolescence management strategy.	Inspection	false
28	ANGARS-122	Lifecycle Cost Estimates	N13 Lifecycle Support	Qualitative	The system shall provide lifecycle cost estimates for sustainment.	Analysis	false
29	ANGARS-123	Maintenance Training	N13 Lifecycle Support	Qualitative	The system shall include training simulators for maintenance crews.	Demonstration	false
30	ANGARS-124	Repair Documentation	N13 Lifecycle Support	Qualitative	The system shall document all repair procedures for field use.	Inspection	false
31	ANGARS-138	Tanker Integration Size/Weight	N14 Integration with Tanker Platforms	Quantitative	The system shall not exceed 750 kg (T), 500 kg (O) or 3 cubic meters (T), 2 cubic meters (O) in size/weight during tanker integration.	Inspection	false

2.5 Key Performance Parameters (KPPs)

Table 8: ANGARS Key Performance Parameters (KPPs)

#	Name	Derived From	classification	Text	Verify Method	KPP
2	ANGARS-2 Refueling Time	N1 Autonomous Refueling	Quantitative	The system shall complete refueling within 10 minutes (T), 8 minutes (O) of handshake (alignment) with receiving aircraft.	Test	true
27	ANGARS-27 Contested Positioning Accuracy	N3 Contested Environment Operation	Quantitative	The system shall maintain positional accuracy of ≤ 5m (T), 4m (O) in contested environments.	Test	true
31	ANGARS-31 Low-Latency Commands	N4 Secure Communication	Quantitative	The system shall maintain ≤ 15ms (T), 10ms (O) latency in command transmission.	Test	true
43	ANGARS-43 AES-256 Encryption	N3 Contested Environment Operation N4 Secure Communication N10 Threat Detection & Response	Binary	The system shall encrypt all communications using AES-256 encryption protocol.	Inspection	true
44	ANGARS-44 Obstacle Detection Range	N5 Collision Avoidance	Quantitative	The system shall detect obstacles within 500m (T), 750m (O) with 99% accuracy.	Test	true
53	ANGARS-53 Collision Expectancy	N5 Collision Avoidance	Quantitative	The system shall have on average, a collision expectancy of no more than once every 1,000,000 mission hours (T) 1,500,000 (O) mission hours.	Analysis	true
56	ANGARS-56 Triple Redundancy	N6 Redundant Fail-Safe Systems	Binary	The system shall employ triple-redundant flight control systems.	Inspection	true
62	ANGARS-62 Mission Success Rate	N7 Mission Reliability	Quantitative	The system shall achieve ≥95% mission success rate under nominal conditions.	Analysis	true
91	ANGARS-91 Jamming Mitigation	N10 Threat Detection & Response	Quantitative	The system shall mitigate jamming attempts within 30 seconds (T), 15 seconds (O).	Test	true
125	ANGARS-125 KC-46 Integration	N14 Integration with Tanker Platforms	Binary	The system shall integrate with existing KC-46 tankers without structural modifications.	Inspection	true

3 ANGARS CONOPS

3.1 CONOPS Derivation Process

The CONOPS of the system defines *how* the system is intended to be used**.** Similar to the Requirements Derivation Process, CONOPS derivation follows a similarly systematic, iterative approach starting with the needs described earlier. Analyzing the needs defined by stakeholders, we will decompose a set of users/operators. It is important that we begin with a scenario-agnostic frame of understanding first so that we can define a full scope of understanding and then decompose that into smaller, more scenario-specific chunks.

Next, we can analyze the interaction between the user/operator and the system's needs to determine and define the interaction, providing a description of the intended operation. We could potentially use this to refine our External Interface Requirements.

Finally, once a holistic definition of the intended system operation is defined, we can start decomposing system operation into scenario-specific operational scenarios, which provide further insight into actions that the system must execute to help refine Functional Requirements.

3.2 OV-1 Concept Diagram

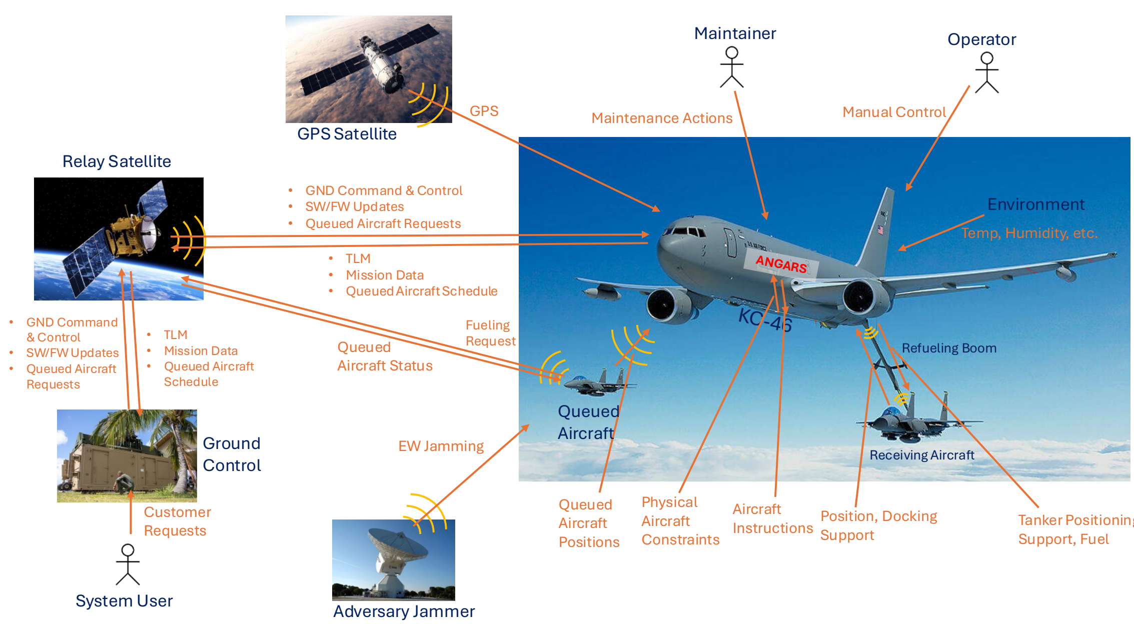


Figure 1: ANGARS OV-1 Concept Diagram

The ANGARS OV-1 displays the expected scope of the intended operation of the system. Starting at the System User icon in the bottom left, the customer will submit a request to a government-defined, owned, and operated Ground Control system, which will then pass those instructions via Satellite to the ANGARS. The relay mechanism is necessary as the ANGARS system can be deployed in a wide range of environments and spaces, allowing for flexibility in the communication chain between ANGARS and the ground. The system will decompose this data into aircraft-executable actions once the mission scenario has been transmitted to ANGARS. This information will then be passed to the aircraft/aircraft operator to direct the aircraft to the mission area.

Once the aircraft is within its mission operation area, it will begin listening to, logging, and evaluating Refueling Requests from aircraft within the area. Once a receiving aircraft is chosen, ANGARS and the receiving aircraft begin the docking sequence. During docking, ANGARS and the aircraft continually communicate via numerous mediums to convey critical docking information. Following successful docking, fueling will begin.

During fueling, ANGARS continually communicates with the ground and the receiving aircraft to ensure that the mission is performing nominally, taking into various account such as fuel level, schedule, and specific receiving aircraft properties.

After fueling has completed, the two aircraft will carefully depart from one another, a mission report will be generated and sent down to the ground, and the ANGARS system will revisit the schedule to determine the next aircraft in the queue.

To further illustrate the role of ANGARS and the stakeholders that it interacts with a Contextual Block Definition Diagram has been provided.

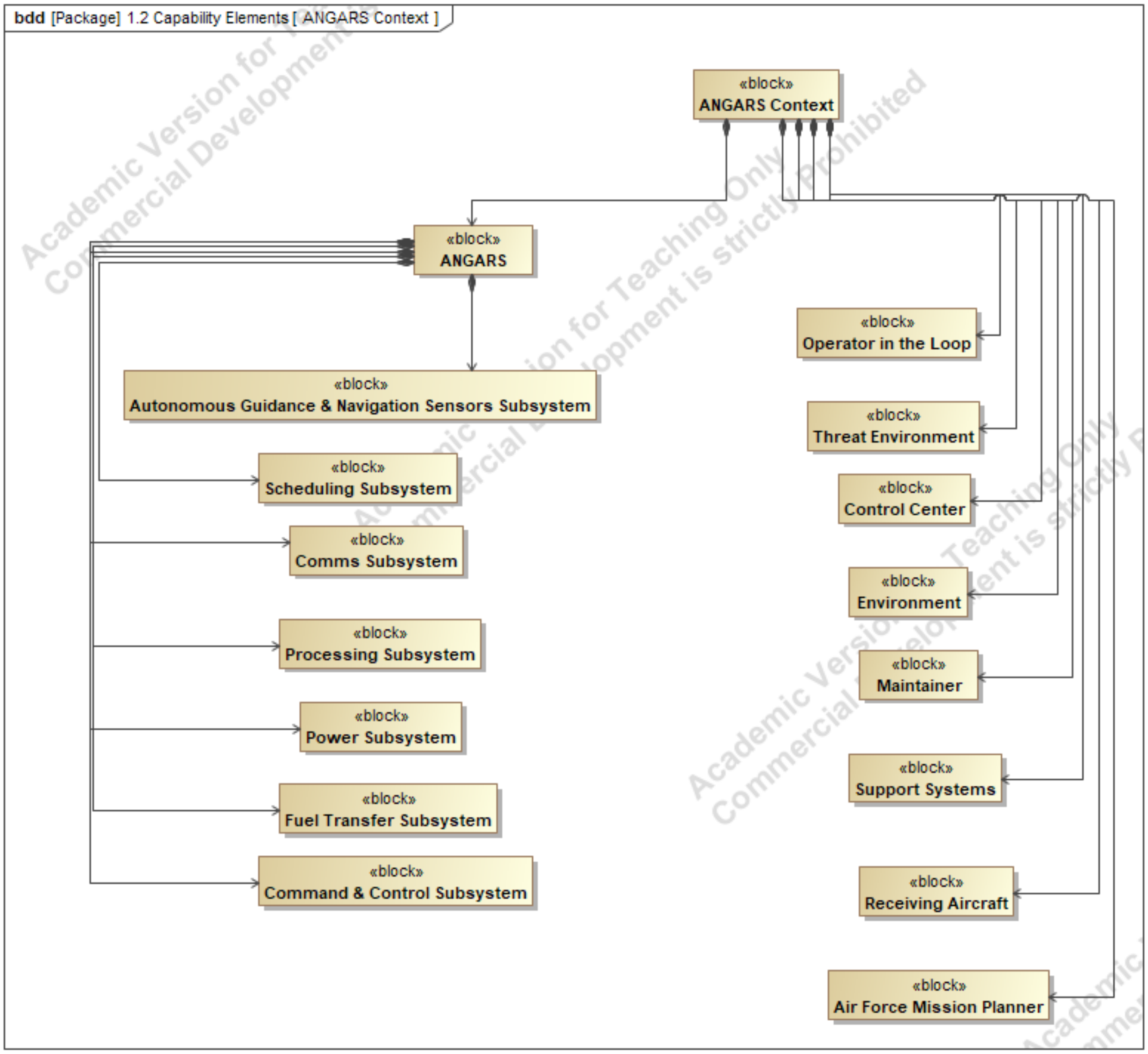


Figure 2: ANGARS Context BDD

3.3 Operational Scenarios

Scenario 1: Refueling Scheduling and Prioritization

Context: ANGARS enters the mission area and starts accepting communication links

Description:

Once other aircraft in the area get notification that an ANGARS tanker is in the area, they will begin attempting to open communication channels with ANGARS. Once a communications data link is established, the requesting aircraft submits a request to refuel through the ANGARS scheduling and management processing system.

When a request is eventually received from a Requesting Aircraft, ANGARS will add this request along with necessary properties of the request, such as priority, time, telemetry, aircraft type, etc. to a scheduler where ANGARS can host multiple requests from a diverse range of requestors. During this phase, ANGARS will continually receive requests and evaluate priorities until the system chooses the next refueling receiver. Priorities are chosen by running properties of aircraft requests against algorithms to determine criticality, timeliness, and efficiency, ultimately resulting in an ordered list of aircraft to be refueled.

Once a receiver is chosen, ANGARS will send messages to all aircraft to inform them of the current schedule and their position in it.

Scenario 2: Autonomous Docking and Fueling

Context: ANGARS and receiving aircraft coordinate and agree on docking with one another.

Description:

The Docking and Fueling phase can be grouped into three substages: meet, greet, and handshake.

The first substage, meet, kicks off as soon as ANGARS and the receiving aircraft have coordinated refueling. The two aircraft will begin navigating toward one another, primarily using GPS and sensors that work effectively at a far range.

The next substage, greet, initiates when the ANGARS tanker and receiving aircraft are within a couple of boom lengths away. During this transition, additional sensors are activated to ensure that the receiving aircraft and ANGARS are aligned and positioned for the boom to be put into position.

Finally, the last substage is once the receiving aircraft is close enough to the ANGARS tanker aircraft that the positioning and sensing system must switch into a different mode of operation to initiate docking/contact. A different set of sensors are used, which can sense finer movements at a higher rate. Throughout this phase, ANGARS and the aircraft will continually communicate via numerous mediums to convey critical docking information such as heading, speed, distance, etc. along with a wide array of sensors working in parallel to coordinate successful docking. The autonomous approach requires the use of multiple sensors and data links to facilitate a successful approach.

Scenario 3: Operation in GPS and EW Impacted Area

Context: ANGARS operates in a GPS-denied environment with active electronic warfare threats. Refueling is not actively occurring at this time.

Description:

The scenario is kicked off when ANGARS detects GPS signal loss or spoofing attempts via onboard threat detection. At the same time, encrypted datalinks are identified to be subject to EW activity, such as jamming attempts. Following these detections, the system autonomously transitions into GPS-denied mode, initializing and activating redundant navigation systems such as terrain-based navigation modes. Sensors are also changed to adapt to the EW controlled environment by prioritizing data from alternative sensor sources (optical, LIDAR, etc.) and other backup navigation methods.

Communication channels are assessed and ANGARS attempts to reroute mission data/comms with the ground by attempting to communicate over various satellite relay networks to regain and establish secure communications with the ground. Once a safe and secure communication network path is established to the ground ANGARS will report critical mission data to provide context and information about the situation.

Information about system response, user inputs, and any situational awareness data is stored and encrypted for review and will become available for ground station to request following EW/GPS jamming attempts. All information systems and live situation data is available to operators on board via the HMI.

3.4 Users & Operators

Users

The system's primary users will be Air Force/Navy pilots and Air Force/Navy mission planners. The ANGARS system will assist Air Force/Navy pilots to maintain sufficient fuel within their aircraft for operation. Air Force/Navy generals can dictate the when, where, and why the ANGARS will be deployed, and Air Force/Navy pilots can submit requests to the ANGARS automated fueling rescheduling system (AFRS) to request fueling services

Operators

Operators are responsible for stepping in and taking over manual commanding in the event that a situation is severe enough to warrant overriding the autonomous system.

Maintainers

Maintainers are responsible for applying non-autonomous maintenance operations to the ANGARS system.

Appendix A – Schedule and EVM Updates

Schedule

Table 9: Capstone Schedule

WBS ID	Task Name	Planned			Actual		
		Work Hours	Milestone Start	Milestone Finish	Work Hours	Milestone Start	Milestone Finish
1	**Project Concept & Proposal**	50	10-Dec	21-Jan	25	10-Dec	1/19/25
1.1	Intro	5					
1.3	System Need	5					
1.4	System Description	15					
1.5	Background	10					
1.6	Justification	5					
1.7	Draft Cleanup & Submittal	5					
1.8	Final Revisions & Submittal	5					
2	**Requirements Analysis Report (RAR) & CONOPS**	45	21-Jan	2-Feb			
2.1	Requirements Definition & Description	8					
2.2	Requirements Verification Plan	6					
2.3	User/Stakeholder Needs Anaylsis & Report	7					
2.4	Key Performance Parameters (KPP) Identification & Compilation	6					
2.5	CONOPS - Concept Diagram	6					
2.6	CONOPS - Potential Users/Operators List	4					
2.7	CONOPS - Use Case Scenario Definition	5					
2.8	RAR & CONOPS Cleanup & Submittal	3					
3	**Functional Analysis Report (FAR)**	37	2-Feb	16-Feb			
3.1	Functional Tree Definition	5					
3.2	Functional Decomp / Tree	5					
3.3	Context Diagram UPDATES	3					
3.4	Function List Development	4					
3.5	Functional Flow Diagrams Creation	6					
3.6	N2 Diagrams Development	5					
3.7	Requirements-Functions Traceability Matrix	4					
3.8	System States & Modes Definition	3					
3.9	FAR Cleanup & Submittal	2					
4	**Trade Study**	32	16-Feb	23-Feb			
4.1	Trade Study Purpose Definition	3					
4.2	Selection Criteria Development	4					
4.3	Alternatives Analysis	5					
4.4	Criteria Weights Development	4					
4.5	Utility Functions Development	4					
4.6	Raw Criteria Values Collection	4					
4.7	Weighted Utility Calculations	3					
4.8	Sensitivity Analysis	3					
4.9	Trade Study Cleanup & Submittal	2					

WBS ID	Task Name	Planned			Actual		
		Work Hours	Milestone Start	Milestone Finish	Work Hours	Milestone Start	Milestone Finish
5	**Concept Design Report**	37	23-Feb	2-Mar			
5.1	Physical Context Diagram Development	5					
5.2	Physical Component Tree Creation	4					
5.3	Subsystem Descriptions	5					
5.4	Physical Block Diagrams Development	6					
5.5	Physical N2 Diagrams Creation	5					
5.6	Data Flow Diagrams	4					
5.7	Interface Definitions & Descriptions	3					
5.8	Functions-Components Mapping	3					
5.9	Concept Design Cleanup & Submittal	2					
6	**Test Plan**	27	2-Mar	9-Mar			
6.1	Test Objectives Definition	3					
6.2	Requirements-Test Objectives Traceability	4					
6.3	Test Environment Description	3					
6.4	Test Equipment Identification	3					
6.5	Test Subjects & Articles Definition	3					
6.6	Success Criteria Development	3					
6.7	Metrics Definition	3					
6.8	Test Execution Planning	3					
6.9	Test Plan Cleanup & Submittal	2					
7	**A-Specification Report**	22	9-Mar	16-Mar			
7.1	Requirements Update & Organization	4					
7.2	KPP Updates	3					
7.3	Support Requirements Development	3					
7.4	Safety Requirements Development	3					
7.5	HSI Requirements Development	4					
7.6	Requirements-Functions Traceability Update	3					
7.7	A-Spec Cleanup & Submittal	2					
8	**Risk Management Report**	15	16-Mar	30-Mar			
8.1	Risk Definitions Development	2					
8.2	Risk Categorization	2					
8.3	Individual Risk Assessments	3					
8.4	Mitigation Steps Documentation	3					
8.5	Risk Progression Analysis	3					
8.6	Risk Report Cleanup & Submittal	2					
9	**Draft Final Report**	16	30-Mar	6-Apr			
9.1	Executive Summary Development	3					
9.2	Report Sections Integration	4					
9.3	Schedule Assessment & EVM Analysis	3					
9.4	Lessons Learned Documentation	2					
9.5	Next Steps Development	2					
9.6	Draft Final Report Cleanup & Submittal	2					
10	**Final Report & Oral Presentation**	11	6-Apr	25-Apr			
10.1	Final Report Revisions	2					
10.2	Presentation Development	3					
10.3	Presentation Materials Preparation	2					
10.4	Final Report & Presentation Submittal	2					
10.5	Oral Presentation Delivery	2					
		292					

EVM

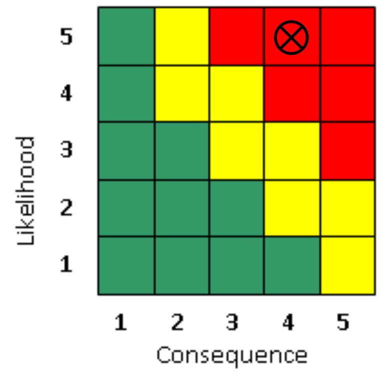
Table 10: EVM Update

					EVM Analysis					
Week	End of Week Date	BCWS	BCWP	ACWP	Cost Variance	Schedule Variance	SPI	CPI	TCPI (BAC)	EAC
1	1/26/25	8	50	5	45	42	6.25	10.00	0.84	29.2
2	2/2/25	26	50	30	20	24	1.92	1.67	0.92	175.2

Appendix B – Risk Updates

Risk 1 – Navigation System Failure

Risk Title	Navigation System Failure	
Description:	IF the GPS integration fails to meet accuracy requirements THEN autonomous positioning during refueling will be compromised, leading to unsafe operating conditions.	
Initial Assessment:	Likelihood:	5
	Consequences:	4
	Description of Consequences if realized	- Autonomous positioning failures would make it difficult to establish rough positioning of the tanker aircraft. Putting the mission, crew members, and others aboard at risk



Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	5	4	Initial assessment
2	RAR	Impose requirements for implementation of backup navigation system. See: ANGARS-22, Inertial Navigation.	5	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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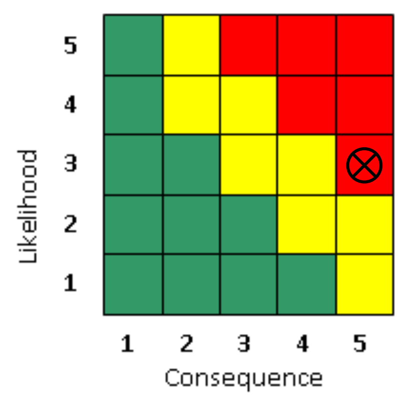
Risk 2 – Communications System Failure

Risk Title	Communication System Failure	
Description:	IF secure data link is lost during operation THEN operational delays or mission failure may occur, potentially requiring manual intervention.	
Initial Assessment:	Likelihood:	3
	Consequences:	5
	Description of Consequences if realized	- Comms going down would result in mission failures. Potential critical situational awareness updates may not be conveyed between ANGARS to ground or receiving aircraft, or vise-versa.



Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	3	5	Initial assessment

2	RAR	Impose requirements defining design-level communication up-time constraints. See: ANGARS-33, Communication Uptime.	3	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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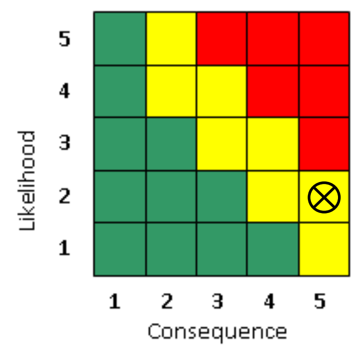
Risk 3 – Collision Avoidance System Failure

Risk Title	**Collision Avoidance System Failure**	
Description:	IF autonomous collision avoidance fails to activate when needed THEN there is an increased risk of mid-air collisions.	
Initial Assessment:	Likelihood:	2
	Consequences:	5
	Description of Consequences if realized	- Autonomous collision avoidance system failures could result in catastrophic consequences such as bird strikes and fuel boom strikes.



Mitigation Plan			**L**	**C**	**Impact Description & Rationale**
ID	Document	Mitigation Action			
1	Proposal	--	2	5	Initial assessment
2	RAR	Impose requirements defining design-level collision expectations. See: ANGARS-53, Collision Expectancy	2	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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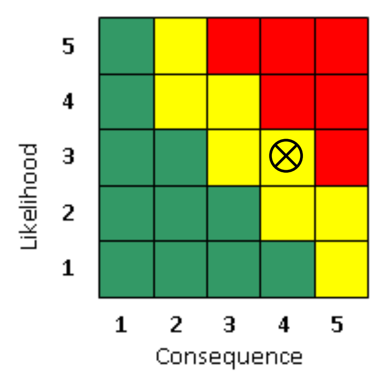
Risk 4 – Integration Challenges

Risk Title	**Collision Avoidance System Failure**	
Description:	IF there are delays in integration with KC-46/NGAS platforms THEN schedule overruns and increased costs will impact project delivery	
Initial Assessment:	Likelihood:	3
	Consequences:	4
	Description of Consequences if realized	- Integration issues could result in significant schedule and cost risks, including overrun/budget. - Potential implications to existing ANGARS architecture or modifications to legacy hardware.



Mitigation Plan			**L**	**C**	**Impact Description & Rationale**
ID	Document	Mitigation Action			

1	Proposal	--	3	4	Initial assessment
2	RAR	Impose requirements defining design-level integration expectations. See: ANGARS-125, KC-46 Integration	3	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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Appendix C - References

KC-46 refueling image: https://skiesmag.com/news/boeing-bid-kc-46-future-rcaf-tanker-program/

Ground Satellite Image: https://www.esa.int/ESA_Multimedia/Images/2005/06/Cebreros_station

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